PANCHANAND JHA

Senior Software Development Engineer in Test (SDET) | QA Automation Engineer
India | Open to Relocate
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PROFESSIONAL SUMMARY

Senior Software Development Engineer in Test (SDET) with **7+ years of experience** designing, developing and maintaining scalable **UI, API and end-to-end test automation** for **Disney+, Disney Streaming Services, Hotstar and ESPN**. Strong hands-on expertise in **WebDriverIO, Selenium, Robot Framework, Playwright and XCUITest**, with programming experience in **TypeScript, JavaScript, Python and SQL**. Experienced in building reusable automation frameworks, regression suites, API validation, integration testing and **GitLab CI/CD** pipelines. Additional experience in **AWS, Docker, AI/ML, Hugging Face Transformers, LLMs and computer vision**, with a strong interest in cloud-native automation, DevOps and intelligent quality engineering.

CORE TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Swift, R

Test Automation: WebDriverIO, Selenium WebDriver, Robot Framework, Playwright, XCUITest

Testing: UI, API, E2E, Integration, Regression, Functional, Smoke, Release Testing

Framework Design: Page Object Model (POM), Data-Driven Testing, Keyword-Driven Testing, Reusable Automation Frameworks

API: REST API Testing, API Validation, Postman, Bruno

CI/CD & Tools: GitLab CI/CD, Git, GitHub, Jira, Agile/Scrum

Cloud: AWS, S3, Lambda, Step Functions

Databases: MySQL, MongoDB

AI/ML: Claude, Krio, AmazonQ integration for automation suite, PyTorch, TensorFlow, Hugging Face Transformers, LLMs, OpenCV, Computer Vision

Web/Other: React, HTML, CSS, Node-red, IoT

PROFESSIONAL EXPERIENCE

ZUCITECH SOFTWARE SOLUTIONS PVT. LTD. | 2020 – Present

Senior Software Development Engineer in Test (SDET) / QA Automation Engineer

Develop and maintain automation solutions across multiple Disney and ESPN platforms, supporting UI, API, integration and end-to-end testing, regression execution and production releases.

SELECTED PROJECTS

Disney+ — Artwork & Metadata Packager
JavaScript, Selenium, WebDriverIO, GitLab, Jira, Cucumber, Jambit

- Developed reusable UI automation components and integrated regression execution with GitLab CI/CD.

Disney+ — Falcon / TV Avails Platform
JavaScript, Selenium, WebDriverIO, GitLab, Jira, Cucumber, Jambit

- Supported **20+ production releases** through automated regression and release validation.
- Developed reusable TypeScript/JavaScript automation components.

Hotstar — Louise Falcon Integration
JavaScript, Selenium, WebDriverIO, GitLab, Jira, Cucumber, Jambit, API Testing

- Developed UI, API and integration automation for content management workflows.
- Validated interactions between content platforms and downstream services.

Disney+ — Acquisition & Distribution / Screeners Re-platform
JavaScript, Selenium, WebDriverIO, GitLab, Jira, Cucumber, Jambit, REST API

- Developed UI/API automation for content acquisition, distribution, title processing and OTT workflows.

ESPN Fantasy
Python, Robot Framework, Selenium, REST API

- Developed UI and REST API automation using Python, Robot Framework and Selenium.

Disney+ — Title Registration / Ensemble & Theatre Management
Selenium, WebDriverIO, JavaScript, TypeScript, REST API

- Developed UI and API automation for title registration, metadata management and theatre/title association workflows.

AI & SOFTWARE ENGINEERING PROJECTS

AI-Based Defect Similarity & Classification

- Developed a semantic defect similarity system using a fine-tuned **Hugging Face Sentence Transformer**. Implemented transformer-based NLP workflows to identify potentially duplicate or related software defects.

AI-Enabled Transport Billing System

- Integrated **Whisper and Ollama-based AI** capabilities for automated transport bill generation. Developed API, database and React-based application components.

Additional AI/ML Work

- Machine-learning-based agricultural commodity price forecasting.
- Named Entity Recognition for medical journal data.
- Mortality risk analysis using Python and Tableau.
- Computer vision and machine-learning applications for robotics and intelligent automation.

EDUCATION

Ph.D. — Industrial Design, National Institute of Technology, Rourkela, India | 2011 – 2016
Research: Machine Learning and AI for Robot Kinematics

M.Tech. — Mechanical Engineering, National Institute of Technology, Rourkela, India | 2007 – 2009

Thesis: Artificial Intelligence Neural Network Application for Prediction of Kinematics of Robot Manipulator

B.Tech. — Industrial & Production Engineering, Guru Ghasidas Central University, India | 2003 – 2007

CERTIFICATIONS

- IBM — Machine Learning with Python, 2021
- Data Science — 360DigiTMG, 2021
- Tableau & Tableau Prep — 360DigiTMG, 2021

RESEARCH & RECOGNITION

- 20+ journal publications, 15+ conference papers, 3 patents and 2 technical books covering AI, Machine Learning, Robotics, Computer Vision and Intelligent Automation.
- Individual Appreciation Award, 2023.
- Research Fellowship — Ministry of Education, Science, Research and Sport of the Slovak Republic.
- CICOPS Research Fellowship — University of Pavia, Italy.